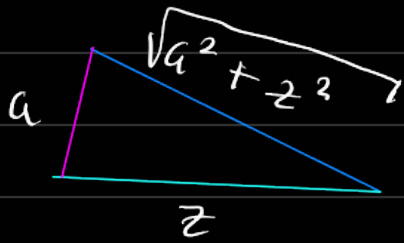
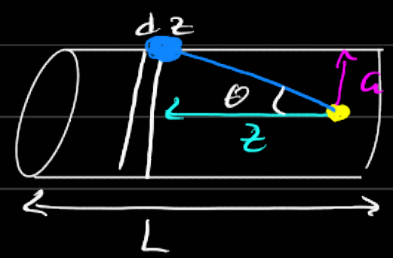


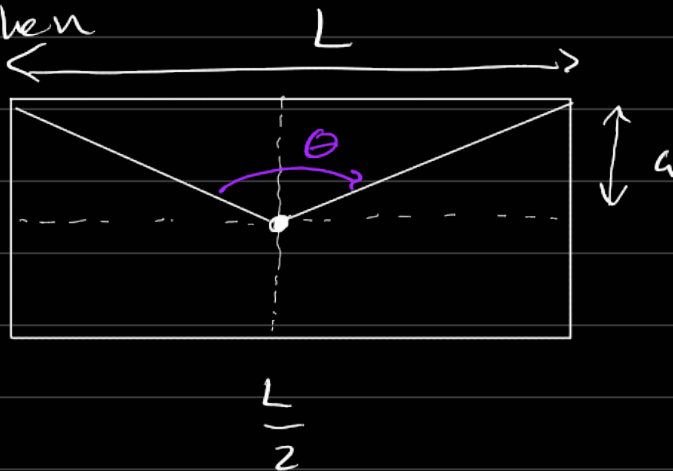
$$B = \int \frac{\mu_0 I a^2}{2(z^2 + a^2)^{3/2}} N_L dz \hat{z}$$

- Point which we take to be the origin
- Point which were measuring B



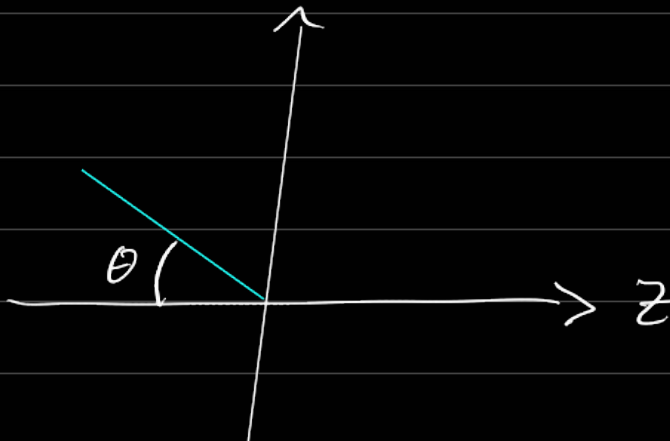
← this is where the $z^2 + a^2$ comes from

* Now, if we want to measure the field at the center, then

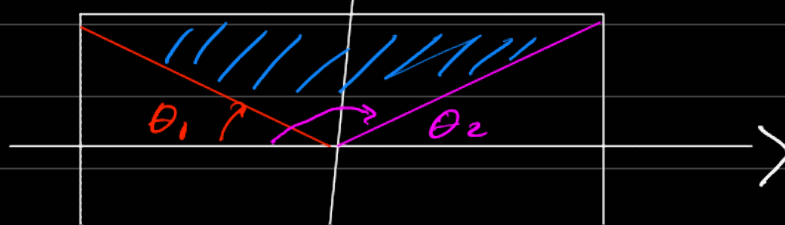


* Only want our θ to be as above

* Note: the way the problem is set up we are measuring θ as complementary angles



Thus, our two bounds are from θ_2 to θ_1



* Using the hint:

$$Z = \frac{a}{\tan(\theta)}$$

but we also need $d\theta$, so we differentiate

$$\frac{dZ}{d\theta} = \frac{d}{d\theta} \left(\frac{a}{\tan(\theta)} \right) = - \left(\frac{a}{\tan^2(\theta)} \right) \sec^2(\theta)$$

↓

$$\frac{d}{d\theta} (\tan(\theta)) = \sec^2(\theta) = \frac{1}{\cos^2(\theta)}$$

$$\begin{aligned} \frac{dZ}{d\theta} &= - \left(\frac{a \cos^2(\theta)}{\sin^2(\theta)} \right) \left(\frac{1}{\cos^2(\theta)} \right) \\ &= - \frac{a}{\sin^2(\theta)} \end{aligned}$$

Thus,

$$dZ = - \frac{a}{\sin^2(\theta)} d\theta$$

$$Z = \frac{a}{\tan(\theta)}$$

* Lets simplify a bit before:

$$Z^2 + a^2 = \left(\frac{a}{\tan(\theta)} \right)^2 + a^2 = a^2 \left(\frac{1}{\tan^2(\theta)} + 1 \right)$$

$$= a^2 \left[\frac{1}{\frac{\sin^2(\theta)}{\cos^2(\theta)}} + 1 \right] = a^2 \left[\frac{\cos^2(\theta)}{\sin^2(\theta)} + 1 \right]$$

$$= \frac{a^2}{\sin^2(\theta)} \left(\cos^2(\theta) + \sin^2(\theta) \right)$$

1

$$= \frac{a^2}{\sin^2(\theta)}$$

Thus

$$\underline{Z^2 + a^2 = \frac{a^2}{\sin^2(\theta)}}$$

* Lets plug this in

$$\vec{B} = \int_{\theta_2}^{\theta_1} \frac{\mu_0 I a^2}{2 (Z^2 + a^2)^{3/2}} N_L dZ \hat{z} = \int_{\theta_2}^{\theta_1} \frac{\mu_0 I a^2 N_L}{2 \left[\frac{a^2}{\sin^2(\theta)} \right]^{3/2}} \left(\frac{-a}{\sin^2(\theta)} d\theta \right) \hat{z}$$

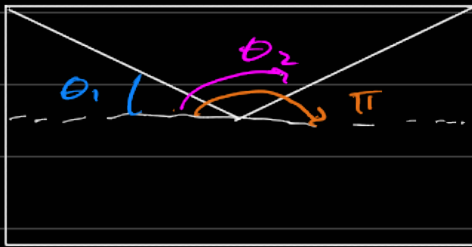
$$= - \int_{\theta_2}^{\theta_1} \frac{\mu_0 I a^3 N_L}{2 a^3} \frac{\sin^3(\theta)}{\sin^2(\theta)} d\theta \hat{z}$$

$$= - \frac{\mu_0 I N_L}{2} \int_{\theta_2}^{\theta_1} \sin(\theta) d\theta \hat{z}$$

Note a has cancelled out.

$$\vec{B} = - \frac{\mu_0 I N_L}{2} \left[-\cos(\theta) \right]_{\theta_2}^{\theta_1} = \frac{\mu_0 I N_L}{2} (\cos(\theta_1) - \cos(\theta_2)) \hat{z}$$

(negatives cancel)



θ_1 and θ_2 are related by
 $\pi - \theta_1 = \theta_2$

trig identity

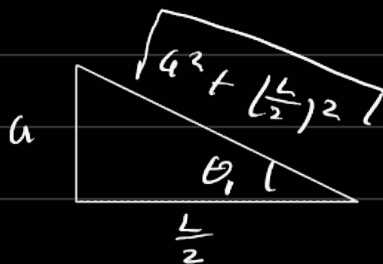
$$\cos(\theta_2) = \cos(\pi - \theta_1) = \cos(\pi)\cos(\theta_1) + \sin(\pi)\sin(\theta_1)$$

$$= -\cos(\theta_1)$$

Plugging this in:

$$\vec{B} = \frac{\mu_0 I N_L}{2} (\cos(\theta_1) - (-\cos(\theta_1))) = \mu_0 I N_L \cos(\theta_1) \hat{z}$$

Finally note



$$\cos(\theta_1) = \frac{L/2}{\sqrt{a^2 + (L/2)^2}}$$

Plugging this in:

$$\vec{B} = \frac{\mu_0 I N_L L}{2} \frac{1}{\sqrt{a^2 + \frac{L^2}{4}}} \hat{z}$$

* For $L \gg a$: factor out $L^2/4$

$$\frac{1}{\sqrt{a^2 + \frac{L^2}{4}}} = \frac{1}{(\frac{L}{2}) \sqrt{a^2 \frac{4}{L^2} + 1}}$$

$$\text{then } \left(\frac{a}{L}\right)^2 \ll 1 \text{ which we can ignore, leaving}$$

$$= \frac{1}{\left(\frac{L}{2}\right)\sqrt{1}} = \frac{1}{\left(\frac{L}{2}\right)}$$

Thus

$$\vec{B}_{(L \gg a)} = \mu_0 I N_L$$

